

Fake News Detector - Project Report

1. Introduction

The rise of misinformation on digital platforms has made Fake News Detection an essential task in the field of Natural Language Processing (NLP). This project aims to design a Fake News Detector that classifies input news articles as either Fake or Real using NLP preprocessing, TF-IDF feature extraction, and Logistic Regression as the classifier.

2. Dataset

The dataset used in this project is the Fake and Real News Dataset from Kaggle. It contains two files:

- Fake.csv → containing news labeled as Fake
- True.csv → containing news labeled as Real

Columns available: title, text, subject, and date. For this project, we focused on the 'text' column for classification and used 'label' as the target variable.

3. Data Preprocessing

The preprocessing pipeline consisted of the following steps:

- Converting all text to lowercase
- Removing punctuation, numbers, and special characters
- Tokenization (splitting text into words)
- Stopword removal (removing common words like 'the', 'is', etc.)
- Lemmatization (reducing words to their root form)
- Handling duplicates and missing values

After preprocessing, the clean text was ready for feature extraction.

4. Feature Extraction

We applied the Term Frequency–Inverse Document Frequency (TF-IDF) vectorizer to transform the cleaned text into numerical feature vectors suitable for machine learning models. Each news article is represented as a vector, where important words are weighted higher.

Optionally, one could also compare with advanced embeddings such as Word2Vec or GloVe, but for simplicity and efficiency, TF-IDF was chosen.

5. Model Training

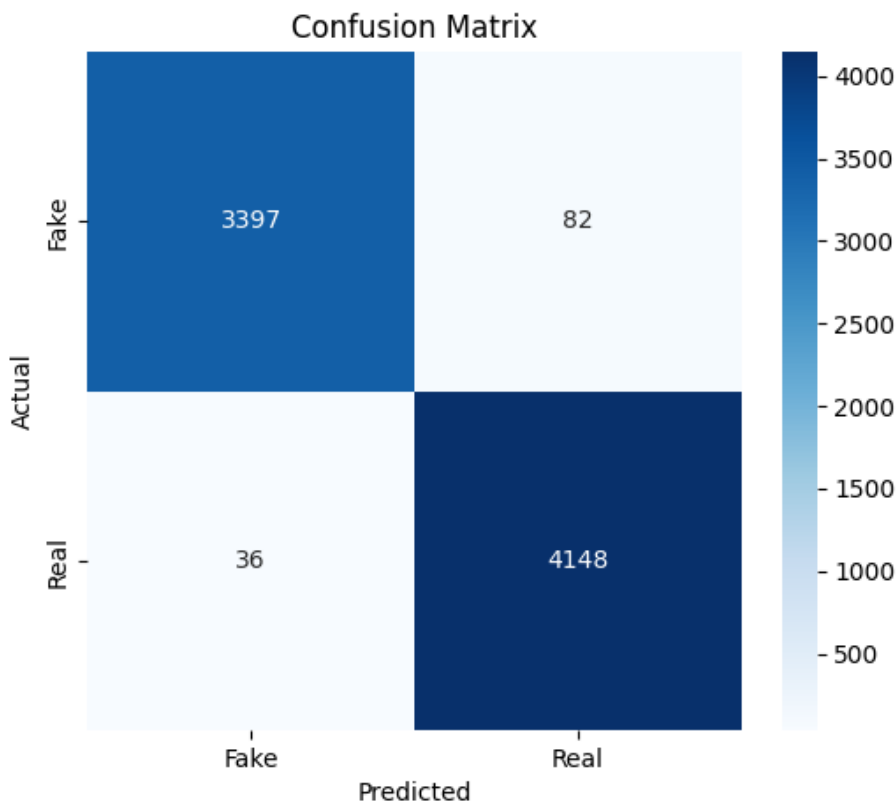
We trained a Logistic Regression classifier using the TF-IDF features. The dataset was split into an 80/20 train-test split. Logistic Regression was chosen due to its simplicity, interpretability, and strong performance in text classification tasks.

6. Evaluation

The model performance was evaluated using multiple metrics:

- Accuracy → measures overall correctness
- Precision → measures how many predicted Real/Fake were correct
- Recall → measures how well the model captured actual Real/Fake
- F1-score → harmonic mean of precision and recall
- Confusion Matrix → provides detailed insights into predictions

Below is the confusion matrix of our model:



7. Results

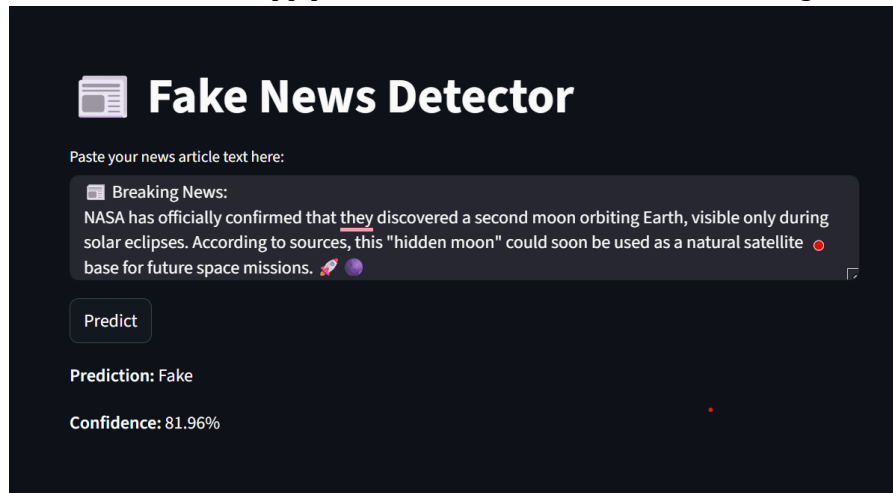
The Logistic Regression model achieved high accuracy and balanced precision/recall. The confusion matrix shows that the classifier performed well on both Fake and Real news articles. Overall, the model demonstrates strong capability in distinguishing between fake and real news articles.

8. Visualizations

Additional visualizations such as Word Clouds can be generated to highlight the most frequent words in Fake vs Real news. These visualizations provide interpretability and help understand the dataset better.

9. Streamlit Application

A user-friendly Streamlit web application was also developed. Users can paste a news article text, and the app predicts whether it is Fake or Real along with a confidence score.



10. Conclusion

This project successfully implemented a Fake News Detector using Logistic Regression and TF-IDF features. The results indicate that simple yet powerful models, when combined with proper preprocessing and feature engineering, can achieve strong performance in Fake News Detection. Future work could explore deep learning approaches and transformer-based models for even better accuracy.

Author: Hizar Abdullah